

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: BenQ
Firmware Version: N/A
Model(s): LX890UST, LW890UST, LH890UST

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.08.0002-b001	1.4.2

Version History

Module Version	Date	Notes
1_0_0_0	8/30/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: InterfaceName.connectionCounter = 5

Supported Classes and Examples

SerialClass
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='LX890UST')
SerialOverEthernetClass
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='LX890UST')
EthernetClass
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 8000, Model='LX890UST')

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value '4:3' '16:10' 'Letterbox' 'Anamorphic 2.35:1'	Value '16:6' 'Auto' 'Wide' 'Anamorphic 16:9'	Value '16:9' 'Real' 'Anamorphic'
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'Computer/YPbPr' 'Component 1' 'DVI-D' 'Composite' 'USB Display' 'DisplayPort'	Value 'Computer 2/YPbPr2' 'Component 2' 'HDMI 1' 'S-Video' 'USB Reader' '3G-SDI'	Value 'Computer 3/YPbPr3' 'DVI-A' 'HDMI 2' 'Network' 'HDBaseT'
# Input example InterfaceName.Set('Input', 'Computer/YPbPr')			
Command LampMode	Value 'Normal' 'Smart Eco 2' 'Custom'	Value 'Eco' 'Smart Eco 3' 'Dual Brightest'	Value 'Smart Eco' 'Dimming' 'Dual Reliable'

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	'Single Alternative'	'Single Alternative Eco'	
# LampMode example InterfaceName.Set('LampMode', 'Normal')			
Command MenuCall	Value 'On'	Value 'Off'	
# MenuCall example InterfaceName.Set('MenuCall', 'On')			
Command MenuNavigation	Value 'Up' 'Right'	Value 'Down' 'Enter'	Value 'Left'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command PictureMode	Value 'Dynamic' 'Bright' 'Cinema' 'Football Bright' 'Silence Mode' 'User 2' 'ISF Night'	Value 'Presentation' 'Living Room' 'Standard/Vivid' 'DICOM' 'DCI-P3 Mode' 'User 3' '3D'	Value 'sRGB' 'Game' 'Football' 'THX' 'User 1' 'ISF Day'
# PictureMode example InterfaceName.Set('PictureMode', 'Dynamic')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command Volume	Value 'Up'	Value 'Down'	
# Volume example InterfaceName.Set('Volume', 'Up')			

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value '4:3' '16:10' 'Letterbox' 'Anamorphic 2.35:1'	Value '16:6' 'Auto' 'Wide' 'Anamorphic 16:9'	Value '16:9' 'Real' 'Anamorphic'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			

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Command Input	Value 'Computer/YPbPr' 'Component 1' 'DVI-D' 'Composite' 'USB Display' 'DisplayPort'	Value 'Computer 2/YPbPr2' 'Component 2' 'HDMI 1' 'S-Video' 'USB Reader' '3G-SDI'	Value 'Computer 3/YPbPr3' 'DVI-A' 'HDMI 2' 'Network' 'HDBaseT'
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampMode	Value 'Normal' 'Smart Eco 2' 'Custom' 'Single Alternative'	Value 'Eco' 'Smart Eco 3' 'Dual Brightest' 'Single Alternative Eco'	Value 'Smart Eco' 'Dimming' 'Dual Reliable'
# LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command LampUsage	Value Hours		
Qualifier Key 'Lamp'	Qualifier Value '1' – '2'		
# LampUsage examples InterfaceName.Update('LampUsage', {'Lamp': '1'}) Value = InterfaceName.ReadStatus('LampUsage', {'Lamp': '1'}) InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command PictureMode	Value 'Dynamic' 'Bright' 'Cinema' 'Football Bright' 'Silence Mode' 'User 2' 'ISF Night'	Value 'Presentation' 'Living Room' 'Standard/Vivid' 'DICOM' 'DCI-P3 Mode' 'User 3' '3D'	Value 'sRGB' 'Game' 'Football' 'THX' 'User 1' 'ISF Day'
# PictureMode examples InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)			
Command	Value	Value	

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Power	'On'	'Off'
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>		
Command	Value	Value
VideoMute	'On'	'Off'
<pre># VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre>		
Command	Value	
VolumeStatus	0 – 10	
<pre># VolumeStatus examples InterfaceName.Update('VolumeStatus') Value = InterfaceName.ReadStatus('VolumeStatus') InterfaceName.SubscribeStatus('VolumeStatus', None, FeedbackHandler)</pre>		

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 115200

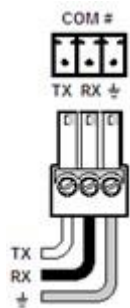
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	8000
Logon Credentials Supported:	No
Multi-Connection	Undetermined
Capabilities:	
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 16:10	\x0D*asp=16:10#\x0D
Aspect Ratio 16:6	\x0D*asp=16:6#\x0D
Aspect Ratio 16:9	\x0D*asp=16:9#\x0D
Aspect Ratio 4:3	\x0D*asp=4:3#\x0D
Aspect Ratio Anamorphic	\x0D*asp=ANAM#\x0D
Aspect Ratio Anamorphic 16:9	\x0D*asp=ANAM16:9#\x0D
Aspect Ratio Anamorphic 2.35:1	\x0D*asp=ANAM2.35:1#\x0D
Aspect Ratio Auto	\x0D*asp=AUTO#\x0D
Aspect Ratio Letterbox	\x0D*asp=LB0X#\x0D
Aspect Ratio Real	\x0D*asp=REAL#\x0D
Aspect Ratio Wide	\x0D*asp=WIDE#\x0D
Audio Mute Off	\x0D*mute=off#\x0D
Audio Mute On	\x0D*mute=on#\x0D
Auto Image None	\x0D*auto#\x0D
Freeze Off	\x0D*freeze=off#\x0D
Freeze On	\x0D*freeze=on#\x0D
Input 3G-SDI	\x0D*sour=SDI#\x0D
Input Component 1	\x0D*sour=YPBR#\x0D
Input Component 2	\x0D*sour=YPBR2#\x0D
Input Composite	\x0D*sour=VID#\x0D
Input Computer 2/YPbPr2	\x0D*sour=RGB2#\x0D
Input Computer 3/YPbPr3	\x0D*sour=RGB3#\x0D
Input Computer/YPbPr	\x0D*sour=RGB#\x0D
Input DisplayPort	\x0D*sour=DP#\x0D
Input DVI-A	\x0D*sour=DVIA#\x0D
Input DVI-D	\x0D*sour=DVID#\x0D
Input HDBaseT	\x0D*sour=HDBASET#\x0D
Input HDMI 1	\x0D*sour=HDMI#\x0D
Input HDMI 2	\x0D*sour=HDMI2#\x0D
Input Network	\x0D*sour=NETWORK#\x0D
Input S-Video	\x0D*sour=SVID#\x0D
Input USB Display	\x0D*sour=USBDISPLAY#\x0D
Input USB Reader	\x0D*sour=USBREADER#\x0D
Lamp Mode Custom	\x0D*lampm=custom#\x0D
Lamp Mode Dimming	\x0D*lampm=dimming#\x0D
Lamp Mode Dual Brightest	\x0D*lampm=dualbr#\x0D

**Global Scripter Module
Communication Sheet**

Lamp Mode Dual Reliable	\x0D*lampm=dualre#\x0D
Lamp Mode Eco	\x0D*lampm=eco#\x0D
Lamp Mode Normal	\x0D*lampm=lnor#\x0D
Lamp Mode Single Alternative	\x0D*lampm=single#\x0D
Lamp Mode Single Alternative Eco	\x0D*lampm=singleeco#\x0D
Lamp Mode Smart Eco	\x0D*lampm=seco#\x0D
Lamp Mode Smart Eco 2	\x0D*lampm=seco2#\x0D
Lamp Mode Smart Eco 3	\x0D*lampm=seco3#\x0D
Menu Call Off	\x0D*menu=off#\x0D
Menu Call On	\x0D*menu=on#\x0D
Menu Navigation Down	\x0D*down#\x0D
Menu Navigation Enter	\x0D*enter#\x0D
Menu Navigation Left	\x0D*left#\x0D
Menu Navigation Right	\x0D*right#\x0D
Menu Navigation Up	\x0D*up#\x0D
Picture Mode 3D	\x0D*appmod=threed#\x0D
Picture Mode Bright	\x0D*appmod=bright#\x0D
Picture Mode Cinema	\x0D*appmod=cine#\x0D
Picture Mode DCI-P3 Mode	\x0D*appmod=dcip3#\x0D
Picture Mode DICOM	\x0D*appmod=dicom#\x0D
Picture Mode Dynamic	\x0D*appmod=dynamic#\x0D
Picture Mode Football	\x0D*appmod=football#\x0D
Picture Mode Football Bright	\x0D*appmod=footballbt#\x0D
Picture Mode Game	\x0D*appmod=game#\x0D
Picture Mode ISF Day	\x0D*appmod=isfday#\x0D
Picture Mode ISF Night	\x0D*appmod=isfnight#\x0D
Picture Mode Living Room	\x0D*appmod=livingroom#\x0D
Picture Mode Presentation	\x0D*appmod=preset#\x0D
Picture Mode Silence Mode	\x0D*appmod=silence#\x0D
Picture Mode sRGB	\x0D*appmod=srgb#\x0D
Picture Mode Standard/Vivid	\x0D*appmod=std#\x0D
Picture Mode THX	\x0D*appmod=thx#\x0D
Picture Mode User 1	\x0D*appmod=user1#\x0D
Picture Mode User 2	\x0D*appmod=user2#\x0D
Picture Mode User 3	\x0D*appmod=user3#\x0D
Power Off	\x0D*pow=off#\x0D
Power On	\x0D*pow=on#\x0D
Video Mute Off	\x0D*blank=off#\x0D
Video Mute On	\x0D*blank=on#\x0D

Global Scripter Module Communication Sheet

Volume Down	\x0D*vol=-#\x0D
Volume Up	\x0D*vol=+#\x0D

Appendix B. Update Commands

Aspect Ratio	\x0D*asp=?#\x0D
Audio Mute	\x0D*mute=?#\x0D
Freeze	\x0D*freeze=?#\x0D
Input	\x0D*sour=?#\x0D
Lamp Mode	\x0D*lampm=?#\x0D
Lamp Usage Lamp 1	\x0D*ltim=?#\x0D
Lamp Usage Lamp 2	\x0D*ltim2=?#\x0D
Picture Mode	\x0D*appmod=?#\x0D
Power	\x0D*pow=?#\x0D
Video Mute	\x0D*blank=?#\x0D
Volume Status	\x0D*vol=?#\x0D